



Francisco Samagaio

Junior Data Engineer
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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Python | SQL | ETL Pipelines | Data Warehousing

Work Experience

Junior Developer (Data Engineer) — Natixis in Portugal

Dec 2024 – Present

- Design and maintain ETL pipelines in EasyMorph and SQL (Apache Hive) for Corporate & Investment Banking teams, transforming multi-source data into the reporting-ready datasets behind ~300 financial reports.
- Migrated ~80 reports from IBM Cognos to EasyMorph, rewriting and optimizing SQL logic and validating outputs against the legacy source.
- Maintain and validate 30+ Hive tables used as the reference source for financing data by teams across the company.
- Implement data quality controls (row-count comparisons, totals reconciliation, outlier detection) to catch anomalies before they reach reports.
- Integrate API-sourced JSON data with internal datasets using SQL and EasyMorph for downstream reporting.

Intern Developer — Natixis in Portugal

Dec 2023 – Dec 2024

- Built an automated Python ETL pipeline (Pandas, NumPy) replacing legacy PowerShell workflows and cutting processing time to one tenth (~90% reduction).
- Ingested and standardized 150+ files (CSV, Excel, TXT, JSON): cleaning, validation and business rules, loading into Oracle and generating Excel reports automatically.
- Ensured data quality with pytest tests, row-count reconciliation and logging of pipeline metrics.
- Used Git, Jenkins and Linux to schedule, run and deploy the pipelines.

Research Intern / Master's Thesis — Fraunhofer Portugal AICOS

Feb 2023 – Jul 2023

- Designed, trained and fine-tuned generative AI models (diffusion models and GANs) in PyTorch for ceramic design and color generation, proposing a novel hybrid architecture combining both approaches.
- Built Python data pipelines to prepare training data and evaluate model outputs, using image quality and color metrics alongside a user study to validate results.

Technical Skills

Programming: Python, SQL, C/C++, Bash

Data Engineering: ETL Pipelines, Data Integration, Data Warehousing, Batch Processing

Data Processing: Pandas, NumPy, Data Cleaning, Data Transformation

Tools & Platforms: Git, Linux, Jupyter, EasyMorph

Databases & Query Engines: Apache Hive, SQL Query Optimization

Foundational Knowledge: PySpark, Medallion Architecture

Currently Learning: Databricks, Apache Airflow, dbt, Docker

Education

University of Porto — Faculty of Engineering | 2018–2023

- Master's Degree in Electrical and Computer Engineering
- Thesis: Generative AI for Ceramic Color Generation

University of Genoa | 2022

- Erasmus+ Exchange Program

Languages

Portuguese — Native · **English** — Fluent · **Spanish** — Intermediate · **French** — Beginner